The following constants/relations are important for the study of electricity:

$e = 1.60 \times 10^{-19}$ C (Fundamental Unit of Charge) $m_p = 1.67 \times 10^{-27}$ kg (Mass of Proton)
$m_e = 9.11 \times 10^{-31}$ kg (Mass of Electron) $K = 8.99 \times 10^9$ Nm^2/C^2 (Electrostatic Constant)
$\epsilon_0 = 8.85 \times 10^{-12}$ C^2/Nm^2 (Permittivity Constant) $\lambda = {dq}/{dl}$ $\eta = {dq}/{dA}$ $\rho = {dq}/{dV}$ $I = {dQ}/{dt}$ $\tau = RC$

1 Problems

1. At a particular instant, a point charge located at the origin is moving at a speed of $\vec{v} = (2.5 \times 10^5)\hat{\imath} + 0\hat{\jmath}$. A *magnetometer*, a device that measures the strength and direction of a magnetic field, is placed at a point P that is 1 meter from the positive x-axis and 2 meters from the positive y-axis.

(a) Without using the cross product, determine the magnitude and direction of the magnetic field generated at the point P at this instant.

(b) Now, using the cross product, determine the magnitude and direction of the magnetic field generated at the point P at this instant.

2. An electron, moving to the right with a velocity of $v = (5 \times 10^4 \ \hat{\imath})\ {}^{m}\!/\!{}_{s}$ finds itself in a parallel-plate capacitor with plates parallel to the x-axis and a magnetic field heading into the page.

   (a) Will the electron be deflected upwards or downwards? ____________

   (b) Assume that the strength of the magnetic field is 1 Tesla. In order for the electron to remain on a straight path through the parallel-plate capacitor, what must the electric field strength be?

   (c) Suppose that tha plates are seperated by 10cm. Using your response from part (b), what must the voltage difference be across the two plates?

3. Similar to how the Earth orbits the Sun according to the Sun's gravitational field, an electron orbits within a magnetic field.

   (a) Using R-T coordinates, if you wanted an electron to move at 10% the speed of light in the z-direction and 15% in the tangential direction. What would the velocity vector look like?

   $\vec{v} =$ ______ $\hat{r} +$ ______ $\hat{t} +$ ______ $\hat{z}$

   (b) If the magnetic field $\vec{B} = (1\hat{\imath} + 1\hat{k})$ T, what would be the radius of orbit for an electron?

(c) How would this change for a proton?

(d) What would be the magnetic field generated by the electron? Ensure you use vector form!

4. **(CHALLENGE PROBLEM).** A very large plate with charge density η is being lifted along the z-axis at a rate of v meters per second. Assume that you have a magnetometer located at the origin and that the plate starts at $z = -100$m.

   (a) Diagram the situation, indicating the charge of the plate and its velocity along with the position of the magnetometer.

   (b) Find an expression for the strength of the magnetic field for $-100m < z < 0m$.

(c) Find an expression for the strength of the magnetic field for $z > 0$.